

# Component Structure and Local Spatial Statistics

How a Disconnected Neighbour Graph Biases Getis–Ord  $G_i^*$  and Largely Spares  
Moran’s  $I$

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## Abstract

**[Placeholder abstract — T-008.]** When neighbours are defined by a  $k$ -nearest-neighbour rule, the resulting neighbour graph can split into two or more pieces that never exchange information. The standard toolkits do warn that the split has occurred, but a warning about occurrence is not a warning about consequence: nothing tells the user what the split does to a hot-spot map. This paper shows that the split biases the Getis–Ord  $G_i^*$  local statistic through a global-standardisation artifact —  $G_i^*$  measures level, and level is exactly what a component split displaces — while leaving Moran’s  $I$  essentially unaffected, because  $I$  measures co-variation, which the split does not displace. Numbers, figures and citations in this draft are placeholders until the evidence tasks land; every one of them will be traceable through `papers/CLAIMS.md`.

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# 1 How to read this paper

This paper is written as lecture notes that happen to contain a new result. It assumes a numerate reader — algebra and reading a graph — and no statistics background at all. Every term of art is defined before it is used, and collected in the glossary in Section 11.

Because the exposition is long by design, each section is tagged with its role, and the tag appears in the table of contents and in the PDF bookmarks. A reader who already knows the background can jump straight to the sections tagged [CONTRIBUTION].

| Section | Role         | What it is for                                           |
|---------|--------------|----------------------------------------------------------|
| 2       | exposition   | Why anyone should care that a neighbour graph can split. |
| 3       | exposition   | Everything statistical the argument needs, from scratch. |
| 4       | exposition   | The failure, stated concretely on a real map.            |
| 5       | contribution | Why $G_i^*$ is disturbed and $I$ is not.                 |
| 6       | evidence     | How often this happens across sample sizes and $k$ .     |
| 7       | contribution | The within-component fix and its effect.                 |
| 8       | contribution | What is new here, against prior work.                    |
| 9       | limitations  | What the claim does not cover.                           |

Table 1: Role of each section. Sections tagged *contribution* carry the new result; the rest can be skipped by a reader who already has the background.

## 2 Introduction [EXPOSITION]

[Placeholder — T-008.] Spatial statistics turns a map into a graph and then asks questions of the graph. The translation step is a modelling choice, and this paper is about a consequence of that choice which the standard tools flag as an event but do not trace through to the map the user actually reads (D-010). The founding statistics are due to ? and ?, with the local framework set out by ?; all three are annotated in Section 10.

Figure 1 shows the object at issue.

## 3 A primer, from first principles [EXPOSITION]

This section is shared verbatim with the companion paper on selective inference, so that each paper can be read on its own. Readers who have read the other paper can skip it.

### 3.1 What this primer assumes

This primer assumes a reader who is comfortable with algebra and with reading a graph, and who has no background in statistics whatsoever. Every term of art is defined before it is used. Terms are set in ***bold italic*** at the point of definition and collected in the glossary at the end of the paper, each with the section number where its definition lives.

PLACEHOLDER: a neighbour graph that has silently split  
component 1

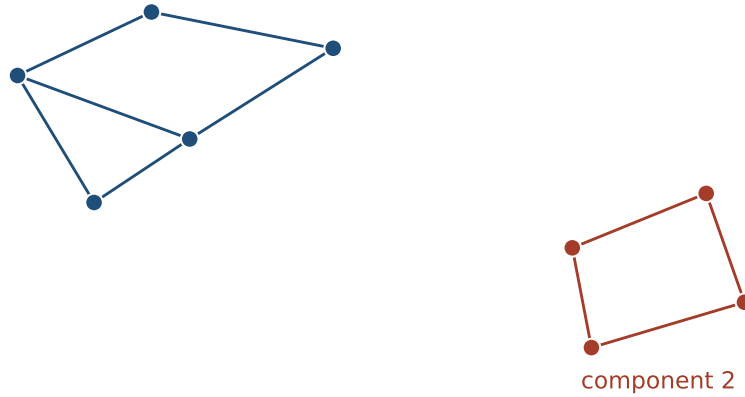


Figure 1: [Placeholder figure.] A  $k$ -nearest-neighbour graph that has split into two components. The toolkits report that this has happened; what they do not report is its effect on  $G_i^*$ . Generated by `papers/shared/mkfigs.py`; replaced by the real figure in the evidence tasks.

### 3.2 Locations, values, and neighbours

Suppose we have measured one number at each of  $n$  places — a rate, a count, a score. Write  $y_i$  for the value at place  $i$ . Spatial statistics asks whether the arrangement of those numbers across the map carries information that the numbers alone do not.

To ask that question we must first say which places count as neighbours of which others. That bookkeeping is done by a *spatial weights matrix*: a square table  $\mathbf{W}$  with one row and one column per place, whose entry  $w_{ij}$  records how strongly place  $j$  counts as a neighbour of place  $i$ . A common choice sets  $w_{ij} = 1$  when  $j$  is one of the  $k$  nearest places to  $i$  and  $w_{ij} = 0$  otherwise.

Reading the table as a map of who-is-connected-to-whom, a group of places that can all reach one another by hopping along nonzero weights, and which no weight connects to anything outside the group, is a *connected component* of the neighbour graph. A neighbour graph with more than one component has been cut into pieces that never exchange information.

### 3.3 Evidence against a hypothesis

Statistical tests work by contradiction. We name a dull explanation for what we observed, then ask how surprising the observation would be if that dull explanation were true. The dull explanation is the *null hypothesis*; here it is usually “the values were scattered across the map with no regard to location.”

The measure of surprise is the *p-value*: the probability of seeing a result at least as extreme as the one observed, computed under the assumption that the null hypothesis holds. A small p-value means the observation sits far out in the tail of what the dull explanation predicts. It does not mean the dull explanation is false, and it is not the probability that the dull explanation is false.

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**Placeholder.** This is placeholder primer content supplied by the build scaffold (T-

005) so that the document compiles, the glossary resolves, and the cross-reference machinery can be checked. T-004 replaces this file with the real from-first-principles exposition required by AC-003.

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## 4 The problem [EXPOSITION]

[Placeholder — T-008.] A concrete statement of the failure on a real map, using the author’s own previously published compute-accessibility atlas as the case study, with the correction stated plainly rather than euphemistically (D-004).

## 5 The mechanism [CONTRIBUTION]

[Placeholder — T-008.] The theoretical core:  $G_i^*$  compares a neighbourhood’s sum against what it would be if the neighbourhood held the *global* mean, standardised by the *global* standard deviation, so it measures level. Moran’s  $I$  measures co-variation. A component split displaces level, not co-variation.

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**Key insight.** The bias is not in  $G_i^*$ ’s arithmetic; it is in the mismatch between a *local* neighbourhood and a *global* standardisation constant, which a disconnected graph makes untenable.

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## 6 Evidence [EVIDENCE]

[Placeholder — T-002 supplies the numbers, T-008 the prose.] Disconnection prevalence across a range of sample sizes and  $k$  values drawn from real geography, reported as the fraction of configurations that disconnect (AC-006), together with the effect on  $G_i^*$  and on  $I$ .

## 7 The correction [CONTRIBUTION]

[Placeholder — T-008.] Standardising within components rather than globally, what it changes, and what it costs.

## 8 What is new here [CONTRIBUTION]

[Placeholder — T-008, informed by the prior-art check in T-001.] This section states the novel contribution and distinguishes it from prior work (AC-008). Where the literature check finds work that partially preempts the claim, the claim is narrowed here rather than the source omitted (D-008).

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**What is new.** Placeholder. The claim stated here must survive the prior-art check, and must be narrowed if it does not.

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## 9 Limitations [LIMITATIONS]

[Placeholder — T-008.] Which claims rest on a single dataset, which are simulation-based, and what observation would falsify them (AC-010). This section also carries the null result required by D-009: Moran’s  $I$  is essentially unaffected by disconnection, and practitioners should keep trusting it.

## 10 Annotated bibliography

The reference list that follows these annotations is generated by BIBTEX from the shared file `shared/refs.bib`. Each entry below states what the source does, why it is credible, and precisely what the present paper takes from it.

- ? *What it does*: introduces the autocorrelation coefficient now universally called Moran’s  $I$ , as a way of detecting whether a quantity measured over a region varies smoothly or erratically in space. *Why it is credible*: the founding paper for the statistic, in *Biometrika*, and the definition in use today is still Moran’s. *What we take*: the definition of the statistic and its ratio form.
- ? *What it does*: defines the family of distance statistics  $G_i$  and  $G_i^*$  for detecting local concentrations of high or low values. *Why it is credible*: the originating paper in *Geographical Analysis*; the statistic is implemented under this definition in every major spatial toolkit. *What we take*: the definition of  $G_i^*$ , including the fact that it standardises against the *global* mean and standard deviation.
- ? *What it does*: sets out the LISA framework, which decomposes a global measure of spatial association into per-location contributions. *Why it is credible*: the standard reference for local spatial statistics, and the basis of the local Moran implementations in `spdep`, `esda` and ArcGIS. *What we take*: the framing of local statistics as decompositions of a global one.

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**Placeholder.** This is a stub annotated bibliography supplied by the build scaffold (T-005) to prove the BIBTEX wiring. T-001 replaces this file, and `refs.bib`, with the retrieved and verified corpus required by AC-004 and AC-005.

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## References

- Luc Anselin. Local indicators of spatial association—LISA. *Geographical Analysis*, 27(2): 93–115, 1995. doi: 10.1111/j.1538-4632.1995.tb00338.x.
- Arthur Getis and J. K. Ord. The analysis of spatial association by use of distance statistics. *Geographical Analysis*, 24(3):189–206, 1992. doi: 10.1111/j.1538-4632.1992.tb00261.x.
- P. A. P. Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23, 1950. doi: 10.2307/2332142.

## 11 Glossary

Every term of art used in this paper, with the section in which it is defined.

**Connected component** A group of places in the neighbour graph that can all reach one another by hopping from neighbour to neighbour, and which is joined to nothing outside the group. A graph with two components has been cut in two. (*Defined in Section 3.2, page 3.*)

**Null hypothesis** The deliberately dull explanation a statistical test tries to contradict — here, that the observed values were scattered across the map with no regard to location. (*Defined in Section 3.3, page 3.*)

**p-value** The probability of seeing a result at least as extreme as the one observed, computed on the assumption that the null hypothesis is true. A small p-value says the observation is unusual under the dull explanation; it is not the probability that the dull explanation is wrong. (*Defined in Section 3.3, page 3.*)

**Spatial weights matrix** A square table with one row and one column per place, whose entry in row  $i$  and column  $j$  records how strongly place  $j$  counts as a neighbour of place  $i$ . It is the object that turns a map into a graph. (*Defined in Section 3.2, page 3.*)